

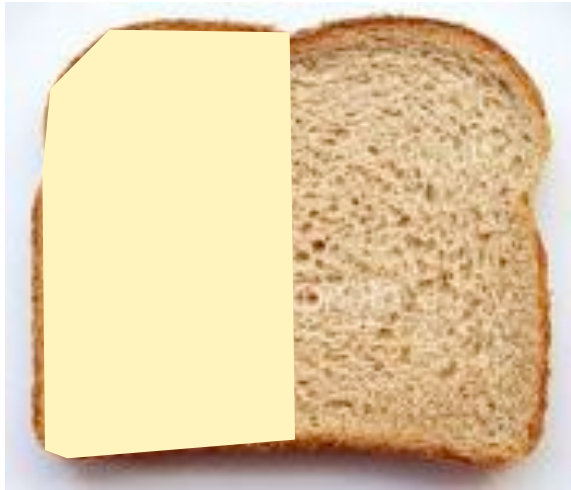
# ***Principal Component Analysis***

## ***Lecture 11***

# *Eigenvectors and Eigenvalues*

---

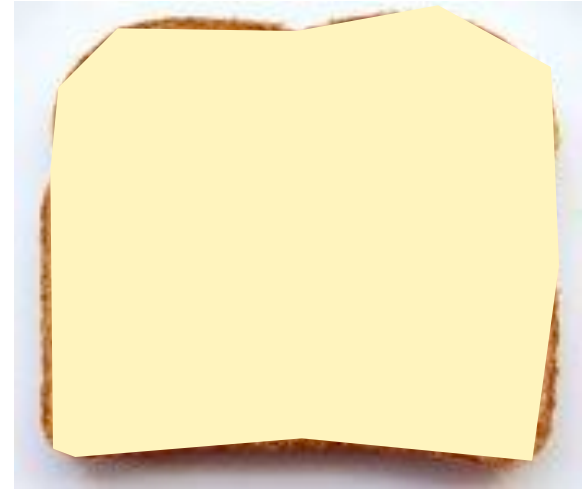
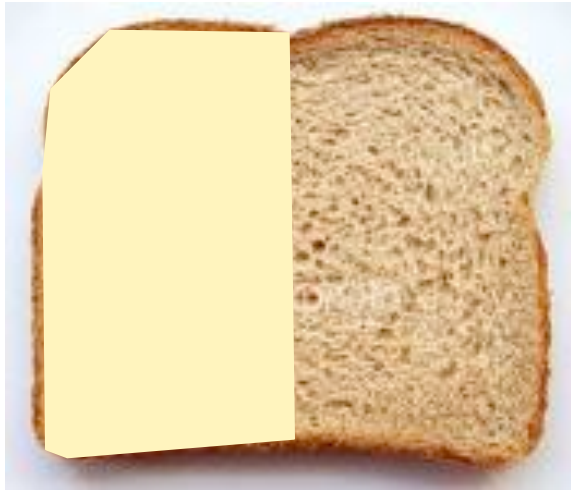
- Consider this problem of spreading butter on a bread slice



# *Eigenvectors and Eigenvalues*

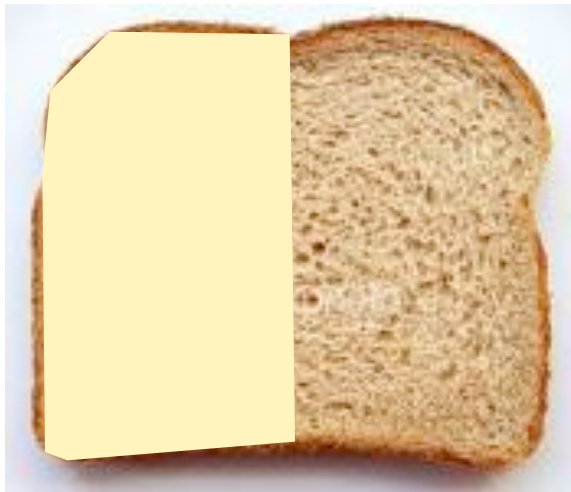
---

- Consider this problem of stretching cheese on a bread slice



# *Eigenvectors and Eigenvalues*

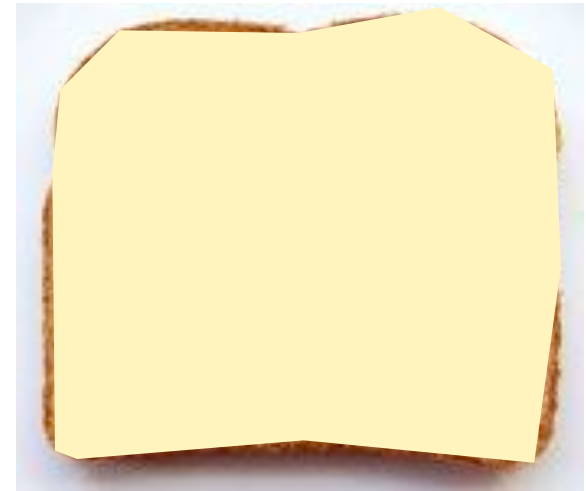
- Consider this problem of spreading butter on a bread slice



$$A = \begin{pmatrix} 2 & 0 \\ 0 & 1 \end{pmatrix}$$

→

*Linear Transformation*



# ***Eigenvectors and Eigenvalues***

---

*Find Eigenvalues:*

$$A = \begin{pmatrix} 2 & 0 \\ 0 & 1 \end{pmatrix}$$

$$|A - \lambda I| = 0$$

$$\begin{pmatrix} 2 - \lambda & 0 \\ 0 & 1 - \lambda \end{pmatrix} = 0$$

$$(2 - \lambda)(1 - \lambda) = 0$$

$$\Rightarrow \lambda = 2, 1$$

# ***Eigenvectors and Eigenvalues***

---

*Find Eigenvectors for  $\lambda=2$*

$$A\vec{v} = \lambda\vec{v}$$

$$(A - \lambda I)\vec{v} = 0$$

$$\begin{pmatrix} 2-2 & 0 \\ 0 & 1-2 \end{pmatrix} \begin{pmatrix} v_1 \\ v_2 \end{pmatrix} = 0$$

$$0v_1 - 1v_2 = 0$$

$$v_2 = 0$$

$$\vec{v} = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$$

*X-axis*

# *Eigenvectors and Eigenvalues*

*Find Eigenvectors for  $\lambda=2$*

$$A\vec{v} = \lambda\vec{v}$$

$$(A - \lambda I)\vec{v} = 0$$

$$\begin{pmatrix} 2-2 & 0 \\ 0 & 1-2 \end{pmatrix} \begin{pmatrix} v_1 \\ v_2 \end{pmatrix} = 0$$

$$0v_1 - 1v_2 = 0$$

$$v_2 = 0$$

$$\vec{v} = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$$

*X-axis*

*Find Eigenvectors for  $\lambda=1$*

$$A\vec{v} = \lambda\vec{v}$$

$$(A - \lambda I)\vec{v} = 0$$

$$\begin{pmatrix} 2-1 & 0 \\ 0 & 1-1 \end{pmatrix} \begin{pmatrix} v_1 \\ v_2 \end{pmatrix} = 0$$

$$1v_1 - 0v_2 = 0$$

$$v_1 = 0$$

$$\vec{v} = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$$

*Y-axis*

# *Eigenvectors and Eigenvalues*

---

## ■ What does the eigenvalue and vectors indicate?

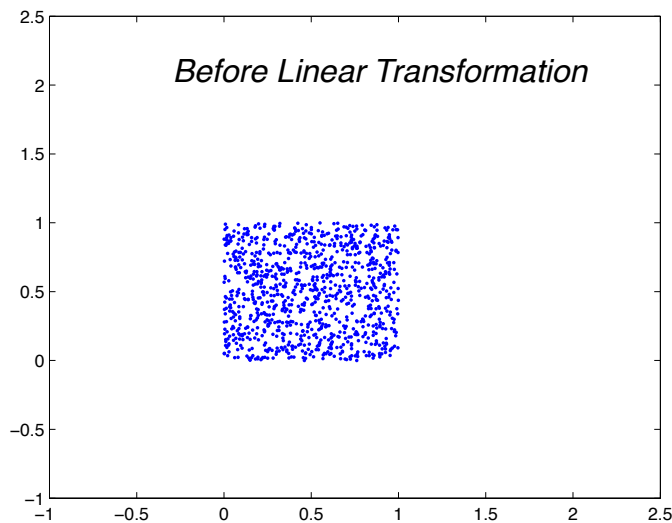
- That the linear transformation results in a scaling of  $\lambda$  along the eigenvector corresponding to  $\lambda$ 
  - In the example presented,  $\lambda=2$  along the x-axis and  $\lambda=1$  along the y-axis (intuitive!)
  - Stated differently, eigenvalue indicates the percentage of transformation present along a particular direction
    - 66.66% along x-axis
    - 33.33% along y-axis

# Eigenvectors and Eigenvalues

## ■ Lets consider another example

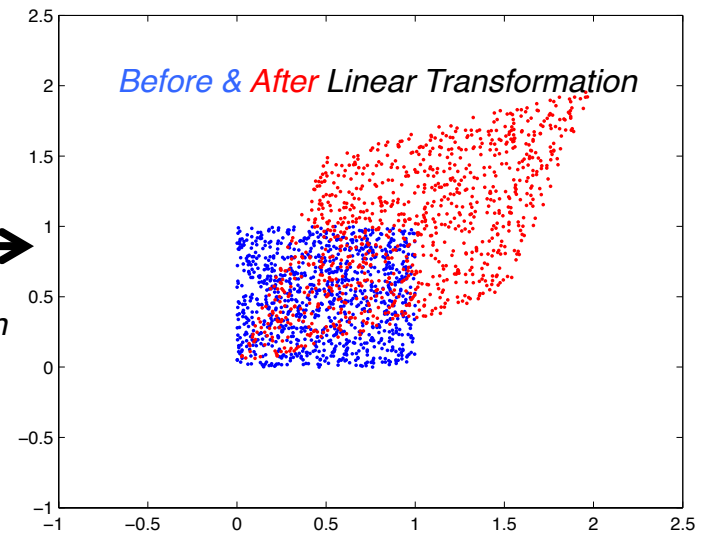
- This time non-zero diagonal elements

$$A = \begin{pmatrix} 1.5 & 0.5 \\ 0.5 & 1.5 \end{pmatrix}$$



$$A = \begin{pmatrix} 1.5 & 0.5 \\ 0.5 & 1.5 \end{pmatrix}$$

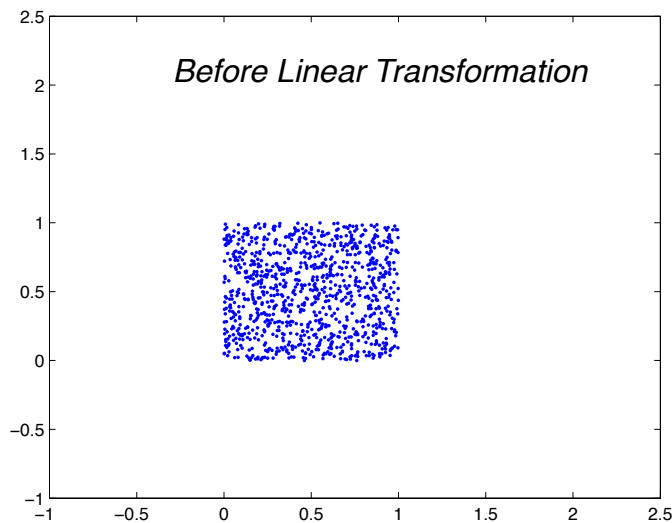
Linear Transformation



# Eigenvectors and Eigenvalues

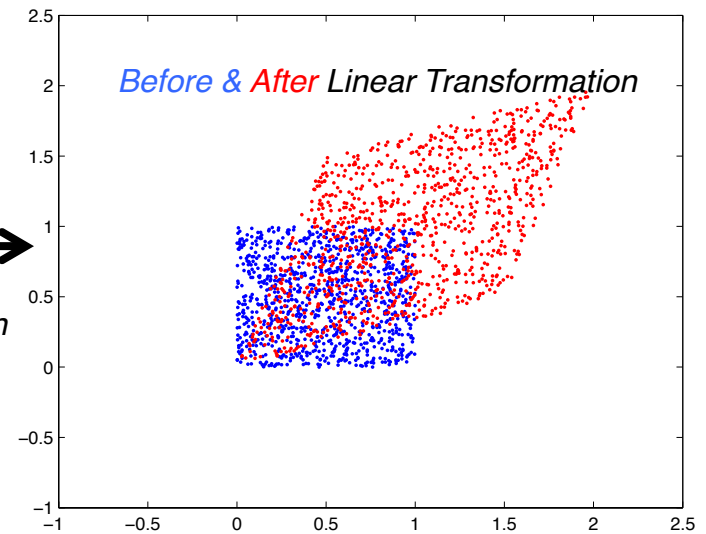
## ■ Compute eigenvalues and eigenvectors of A

$$A = \begin{pmatrix} 1.5 & 0.5 \\ 0.5 & 1.5 \end{pmatrix}$$



$$A = \begin{pmatrix} 1.5 & 0.5 \\ 0.5 & 1.5 \end{pmatrix}$$

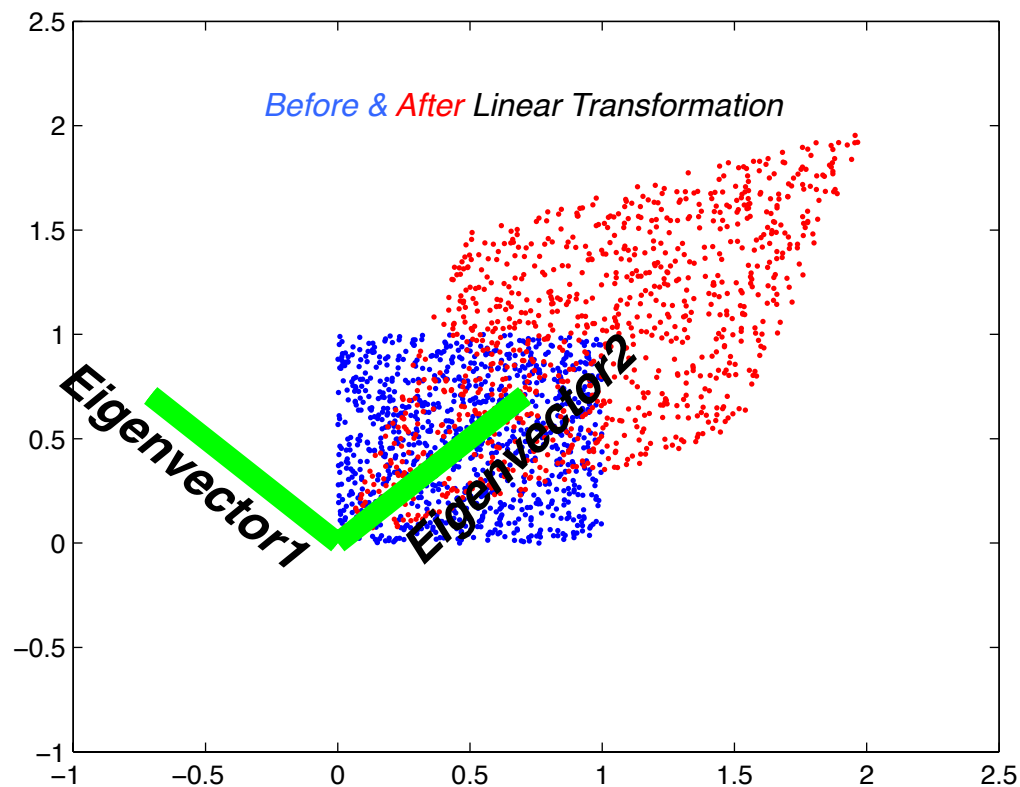
Linear Transformation



# Eigenvectors and Eigenvalues

## ■ Compute eigenvalues and eigenvectors of A

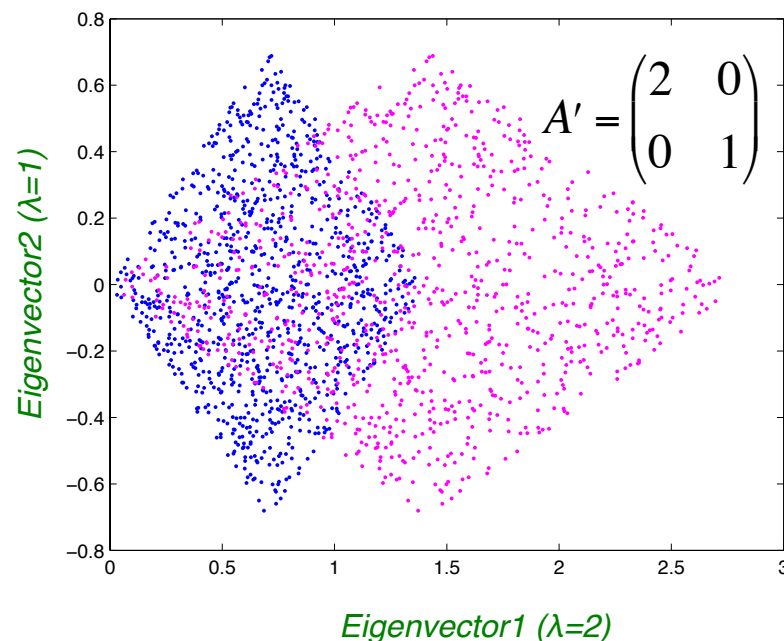
$$A = \begin{pmatrix} 1.5 & 0.5 \\ 0.5 & 1.5 \end{pmatrix} \quad \begin{matrix} \lambda = 1 \\ \vec{v} = \begin{pmatrix} -0.7071 \\ 0.7071 \end{pmatrix} \end{matrix} \quad \begin{matrix} \lambda = 2 \\ \vec{v} = \begin{pmatrix} 0.7071 \\ 0.7071 \end{pmatrix} \end{matrix}$$



# Eigenvectors and Eigenvalues

## ■ What does the eigenvalue and vectors indicate?

- That the linear transformation results in a scaling of  $\lambda$  along the eigenvector corresponding to  $\lambda$ 
  - $\lambda=2$  along  $[0.7071, 0.7071]^T$
  - $\lambda=1$  along  $[-0.7071, 0.7071]^T$



*Even though we started with a non-diagonal transformation matrix ( $A$ ), by computing the eigenvectors and projecting the data onto those eigenvectors allows us to diagonalize the transformation*